```
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>basic</title>
</head>

<body>



    <script>

        // 1. Write a program to check if a number is even or odd.
        function oddEven(num) {
            return num % 2 == 0 ? "odd-number" : "even-number";
        }
        console.log(oddEven(5))

        // Write a program to find the greatest of 3 numbers using if...else
if...else.

        //check a>b then inside  a>c
        //since b is not compared with c
        //check b>c
        function Greatest(a, b, c) {
            if (a > b) {
                if (a > c) {
                    console.log("a is greater" + a);
                }
                else {


                    console.log("c is grater" + c);
                }
            }
            else if (b > c) {
                {
                    console.log("b is grater" + b);
                }

            }
            else {
                console.log("c is grater" + c)
            }

        }
```

```
Greatest(60, 50, 66);

// 3. Create a pattern like this using for loop:

// *

// **

// ***

// ****
var n = 5

for (i = 0; i <= n; i++) //outer will run once and inner will run till
i value
{
    let star = "";
    for (j = 0; j <= i; j++) // this loop count added each time
    {
        star += "*"
    }
    console.log(star);
}



// 4. Print the reverse of a string using a loop.

//      For example:
//          Input: "hello" → Output: "olleh"

// str2=str.split("").reverse().join("")
// console.log(str2)

var s = "hello"
s2 = ""

for (i = s.length; i >= 0; i--) {


    s2 += s.charAt(i)
}
console.log(s2)
```

```
// 5. Write a program to Check whether the number is prime or not?

       var num = 5;

       for (i = 0; i < num; i++)
           if (num % 2 == 0 || num > 2) {
               console.log("the number is not a prime number :" + num);
           }
           else
               console.log("the number is a prime number :" + num);




       // 6. Write a function to generate the following pattern:

       // 1

       // 1 2

       // 1 2 3

       // 1 2 3 4

       var n = 4;

       for (i = 1; i <= n; i++) {

           var empty = "";
           for (j = 1; j<= i; j++) {

               empty += j;

           }
           console.log(empty)
       }
```

```
// 7. Write a program that checks whether a string is a palindrome (same
forward and backward), without using built-in reverse methods.

        str="mada"
        str2=str.split("").reverse().join("")
        if(str==str2) console.log("palindrom")
        else console.log("not apalindrom")


       var string ="madam"
       empty="";
       for(i=string.length; i>=0; i--)
       {
           s2=string.charAt(i)
           empty+=s2

        }
       console.log("reverse of string is "+empty)
       console.log(string === empty ? "the string is palindrome"  : "the
string is not a palindrome");



//         8. Generate Fibonacci series up to N numbers using a while loop.
// 0 1 1 2 3 5 8

     function generateFibonacci(n) {

     let num1 = 0, num2 = 1, count = 0;

  while (count < n) {

    console.log(num1);

    let next = num1 + num2;
    // next= 0+1  1+1  1+3   2+4
    num1 = num2;
    //num1= 1      1     2      4
    num2 = next;
    //num2=1       2     4      6
    count++;

  }
}


generateFibonacci(10);
```

```
// 9. Write a function that takes a number and returns:

// "Fizz" if divisible by 3

// "Buzz" if divisible by 5

// "FizzBuzz" if divisible by both 3 and 5

// The number itself otherwise

function FizzBuzz(num)
{
    if (num%3==0 && num%5==0)
    return "fizzbuzz"
    else if (num%3==0)
    return "fizz"
    else if (num%5==0)
    return "buzz"
    else return num
}
console.log(FizzBuzz(1))
```

```
// 10. Write a function to count how many times each character appears in a
string.
// Input: hello
// Output: { h: 1, e: 1, l: 2, o: 1 }

? Input: hello

? Output: { h: 1, e: 1, l: 2, o: 1 }

function countCharacters(str) {

  const freq = {};

  for (let char of str) {

    if (freq[char]) {

      freq[char]++;

       }
     else {
      freq[char] = 1;
    }
  }

  return freq;
}

console.log(countCharacters("hello"));
```

```javascript
// 11. You have an array of numbers. Return a new array with only the unique elements (no duplicates) using loops and conditionals only.

function getUniqueElements(arr) {
  let uniqueArr = [];

  for (let i = 0; i < arr.length; i++) {
    let exists = false;

    for (let j = 0; j < uniqueArr.length; j++) {
      if (arr[i] === uniqueArr[j]) {
        exists = true;
        break;
      }
    }

    if (!exists) {
      uniqueArr.push(arr[i]);
    }
  }

  return uniqueArr;
}

let numbers = [1, 2, 2, 3, 4, 4, 5];

console.log(getUniqueElements(numbers));
```

```
// arr[i]==arr[j]   =>arr[i]=


let arr=[5,7,5,8,10,12,8]

for(i=0;i<arr.length; i++)
{
    for(j=i+1; j<arr.length; j++)
    {
        if(arr[i]==arr[j])
        {
            arr[i]=""
        }
    }
}
a=[]
for(i=0;i<arr.length;i++)
{
    if(Number.isInteger(arr[i]))
    {

        a[a.length]=arr[i]
    }
}console.log(a)
```

```
// 12. Create a pyramid pattern of stars for a given number of rows:

//       *

//      * *

//     * * *

//    * * * *

num=4;

for(i=0;i<=num;i++)
{
    empty="";

    for (let j = 1; j <= num-i; j++) { // 1st it will leave 1 space and then
execute next for loop
      empty += " ";   //next, time it will give two space and then execute
next loop twice
    }
    for(k=1;k<=i;k++)
    {
     empty+="* "
    }
console.log(empty)
}




// 13. Write a JavaScript program to find factorial of a number using a for
loop.

// 5 x 4 x 3 x 2 x 1 = 120

let fib=4;
// fact=num * (num-1) * (num-2) * (num-3)*(num-4)
// console.log(fact)
var a=1
for (i=1; i<=fib; i++)
{

   a*=i
}
console.log(a)
```

```
// 14. Write a JavaScript program that checks if two strings are anagrams
(e.g., "listen" and "silent").


function areAnagrams(str1, str2) {

  let s1 = str1.replace(/\s/g, "").toLowerCase();
  let s2 = str2.replace(/\s/g, "").toLowerCase();

  if (s1.length !== s2.length) return false;

  let sortedStr1 = s1.split("").sort().join("");
  let sortedStr2 = s2.split("").sort().join("");

  return sortedStr1 === sortedStr2;
}

let word1 = "listen";
let word2 = "silent";

console.log(`Are "${word1}" and "${word2}" anagrams?`, areAnagrams(word1,
word2));

//anagram
var aa='string';
var bb='string';
var a=[...aa];
var b=[...bb];
// function fun(s1,s2){
//     var s1=s1.split('');
//     s1.sort();
//     var s2=s2.split('');
//     s2.sort();  //use s2.join() to make it string and compare without loop
//     if(s1.length!=s2.length)return 'Not an ANAGRAM'
//     else {
//         for (let i = 0; i < s1.length; i++) {
//             if(s1[i]!=s2[i])return 'Not an Anagram'
//         }
//         return a+'\t'+b +' are anagrams';
//     }
// return s1;
// }
// console.log(fun(a,b));

// compare and make one empty ; if empty anagram or else not
```

```
function fun(s1,s2){
    if(s1.length!=s2.length)return 'Not an Anagram'

    for (let i = 0; i <s1.length; i++) {
        for (let j = 0; j <s2.length; j++) {

            if(s1[i]==s2[j]){  //compare to string
                s2[j]=''  //
                break;
            }
        }
    }
    for (let i = 0; i <s2.length; i++) {
        if (s2[i]!='') {
            return 'Not an Anagram'
        }
    }
    return 'Anagram'
}
console.log(fun(a,b));
```

```
// 15. Reverse the words in a sentence, not characters.
// // Input: "hello world from js"

// // Output: "js from world hello"

function reverseWord(sentence)
{
  let word=sentence.split(" ")
  let reverse=word.reverse()
//   return word
//   return sentence
  return reverse.join(" ")

}
console.log(reverseWord("hi how are you"))

// without using inbuilt method
var str='hi how are you';
s1=[''],
count=0; // will pass as the index

for(i=0;i<str.length;i++){

    if(str[i]==' '){  //if empty  - (2) this condition executes
        s1[++count]='' //-count added-initialize the next index with empty
string
        //each character added to the empty string and added as word
    }

    else{            //(1)  this condition executes
        s1[count]+=str[i] //else=str - store as a word in one index
    }
}

output='';
for(i=s1.length-1;i>=0;i--){   // revesrse the word
    output+=s1[i]+' '  //add final with an empty
}

console.log(output)
```

```
// 16. Write a js program to find the second highest number in an array.

arr1=[5,4,28,53,47,98]

for(i=0; i<arr1.length;i++)
{
  for(j=i+1; j<arr1.length;j++)
  {
     if(arr1[i]<arr1[j])
     {
     temp=arr1[i]
     arr1[i]=arr1[j]
     arr1[j]=temp

     }
    }

}
console.log(arr1[1])
// 17. Find Missing Number in a Sequence of numbers from 1 to N

function fms(num)
{
    count=1
    for(i=0; i<num.length; i++)
    {
        if(count!=num[i])   //
        {
            console.log(count)   // print the count
            i-- // to restore the i value - the index value will be same
        }count ++   //increment count after the condition block
    }


}

let numbers = [1, 2, 5,10];
fms(numbers)
```

```
// 18. Write a JS program to Check if the given number is Armstrong Number or
not
 //1³ + 5³ + 3³ = 1 + 125 + 27 = 153 3length of digit
 num=153;
//  a=Math.trunc(Math.log10(num)+1)
 numstr=num.toString();
 pow=numstr.length

 sum=0
 for (i=0; i<pow; i++)
{
    sum+=Math.pow(parseInt(numstr[i]),pow)
}
 if(sum===num)
 {
    console.log("armstrong")

 }
 else{
    console.log("not armstron")
 }

//

let number =154
let strnum=""+number
let final=0
for (let i in strnum )
{
    final+=strnum[i]**strnum.length
}
console.log (final==number ? "armstrong" : "not armstron")


// maximum difference between tw number
```

```
// 19. Sort an Array Without Built-in sort()


  function sortArray(arr) {
  let n = arr.length;

  for (let i = 0; i < n - 1; i++) {

    for (let j = 0; j < n - i - 1; j++) {

      if (arr[j] > arr[j + 1]) {


        let temp = arr[j];

        arr[j] = arr[j + 1];
        arr[j + 1] = temp;
      }
    }
  }

  return arr;
}

let number = [5, 2, 8, 3, 1];
console.log("Sorted array:", sortArray(number));
```

```javascript
// 20. Find the maximum difference between two numbers in an array in
JavaScript?


function maxDifference(arr) {
  if (arr.length < 2)
  return 0;

  let minElement = arr[0];
  let maxDiff = arr[1] - arr[0];

  for (let i = 1; i < arr.length; i++) {

    let currentDiff = arr[i] - minElement;

    maxDiff = Math.max(maxDiff, currentDiff);
    minElement = Math.min(minElement, arr[i]);
  }

  return maxDiff;
}

const arr = [2, 3, 10, 6, 4, 8, 1];

console.log("Maximum difference:", maxDifference(arr));
//-----------------------------------------
arr=[1,2,5,7,8,9,12,6]
let min=arr[0]
let max=arr[0]
for( let i in arr)
{
    if(arr[i]<min) min=arr[i]
    if(arr[i]>max) max=arr[i]
}

let diff=max-min
console.log(diff)
```

```javascript
// 21. Find the largest number in an array in JavaScript.
//declare a variable ->store first index
//for loop
// compare variable with all other element
// asign if the condition is true
function findLargest(arr) {
  let max = arr[0]; // Assume first element is largest
  for (let i = 1; i < arr.length; i++) {
    if (arr[i] > max) {
      max = arr[i]; // Update max if current element is greater
    }
  }
  return max;
}

// 22. How Remove the first element from an array in JavaScript?
// run for loop from i=1
function removeFirstElement(arr) {
  let newArr = [];
  for (let i = 1; i < arr.length; i++) {
    newArr[newArr.length] = arr[i]; // Skip the first element
  }
  return newArr;
}

// 23. Write a Program to find a sum of an array?
//emty=0
//for loop
//empty+=arr[i]

function sumArray(arr) {
  let sum = 0;
  for (let i = 0; i < arr.length; i++) {
    sum += arr[i]; // Add each element to sum
  }
  return sum;
}
```

```javascript
// 24.Write a Program to merge two arrays in JavaScript?
// empty array
// for loop---two array
//empty[empty.length]=arr[i]and a[j]

function mergeArrays(arr1, arr2) {
  let merged = [];

  // Add elements of first array
  for (let i = 0; i < arr1.length; i++) {
    merged[merged.length] = arr1[i];
  }

  // Add elements of second array
  for (let i = 0; i < arr2.length; i++) {
    merged[merged.length] = arr2[i];
  }

  return merged;
}
```

```
// 25. Find the Intersection of Two Arrays in JavaScript?
// check each element----> using 2 for loop ==
//if condition true- store 1st array in empty array
arr=[5,10,52,12,12,20]
arr2=[6,10,52,12,12,23]

let newarr=[]
for (let i=0; i<arr.length ; i++)
{
    for (let j=0; j<arr2.length ; j++)
  {
    if(arr[i]===arr2[j])
    {
        newarr[newarr.length]=arr[i]
    }

}
}
console.log(newarr)
//------------

function intersection(arr1, arr2) {
  let result = [];

  for (let i = 0; i < arr1.length; i++) {
    for (let j = 0; j < arr2.length; j++) {
      if (arr1[i] === arr2[j]) {
        // Check if already added to result


        let alreadyExists = false;

        //remove duplicates
        for (let k = 0; k < result.length; k++) {
          if (result[k] === arr1[i]) {
            alreadyExists = true;
            break;
          }
        }
        if (!alreadyExists) {
          result[result.length] = arr1[i];
        }
      }
    }
  }

  return result;
}
```

```javascript
// 26. Find the Union of Two Arrays in JavaScript?
// we will directly push to result array
// we will declare a var with false
//
let arr=[5,10,52,12,20]
let arr2=[6,10,52,12,23]

function union(arr1, arr2) {
  let result = [];

  // Add all elements from arr1
  for (let i = 0; i < arr1.length; i++) {
    result[result.length] = arr1[i];
  }

  // Add only non-duplicate elements from arr2
  for (let i = 0; i < arr2.length; i++) {

    let isDuplicate = false;
    for (let j = 0; j < result.length; j++) {
      if (arr2[i] === result[j]) { //check whether arr2 value are in result
        isDuplicate = true; // duplicates -if true the loop will cut off
        break; //
      }
    }
    if (!isDuplicate) { //no duplicates -will add
      result[result.length] = arr2[i];
    }
  }

  return result;
}
console.log(union(arr,arr2))

// 27. Write a Program to find the minimum value in an array in JavaScript?
function findMinimum(arr) {
  let min = arr[0];
  for (let i = 1; i < arr.length; i++) {
    if (arr[i] < min) {
      min = arr[i]; // Update if smaller value is found
    }
  }
  return min;
}
```

```javascript
// 28. Check if a String Contains Another String in JavaScript?
function contains(str, sub) {
  for (let i = 0; i <= str.length - sub.length; i++) {
    let match = true;
    for (let j = 0; j < sub.length; j++) {
      if (str[i + j] !== sub[j]) {
        match = false;
        break;
      }
    }
    if (match) return true;
  }
  return false;
}

// 29. Find the First Non-Repeated Character in a String in JavaScript?

function firstNonRepeatedChar(str) {
  for (let i = 0; i < str.length; i++) {

    let count = 0;
    for (let j = 0; j < str.length; j++) {
      if (str[i] === str[j]) count++;
    }
    if (count === 1) return str[i];
  }
  return null;
}
```

```javascript
// 30. Find the Longest Word in a String in JavaScript?

function longestWord(sentence) {
  let word = "";
  let maxLength = 0;
  let current = "";

  for (let i = 0; i <= sentence.length; i++) {
    let char = sentence[i] || " "; // Add space at end for final word

    if (char !== " ") {
      current += char;
    } else {
      if (current.length > maxLength) {
        word = current;
        maxLength = current.length;
      }
      current = "";
    }
  }
  return word;
}

// 31. Capitalize the First Letter of Each Word in a Sentence in JavaScript?

function capitalizeWords(sentence) {
  let result = "";
  let capitalize = true;

  for (let i = 0; i < sentence.length; i++) {
    let char = sentence[i];
    if (char === " ") {
      capitalize = true;
      result += char;
    } else {
      if (capitalize) {
        if (char >= 'a' && char <= 'z') {
          char = String.fromCharCode(char.charCodeAt(0) - 32);
        }
        capitalize = false;
      }
      result += char;
    }
  }
  return result;
}
```

```javascript
// 32. Convert an Array of Strings to Uppercase in JavaScript?

function toUpperCaseArray(arr) {
  let upper = [];

  for (let i = 0; i < arr.length; i++) {
    let str = arr[i];
    let temp = "";

    for (let j = 0; j < str.length; j++) {
      let char = str[j];
      if (char >= 'a' && char <= 'z') {
        char = String.fromCharCode(char.charCodeAt(0) - 32);
      }
      temp += char;
    }
    upper[upper.length] = temp;
  }

  return upper;
}

// 33. Get the last element of an array in JavaScript?

function getLastElement(arr) {
  return arr[arr.length - 1];
}

// 34. Calculate the factorial of a number using recursion in JavaScript?

function factorial(n) {

  if (n === 0 || n === 1) return 1; // Base case
  return n * factorial(n - 1); // Recursive call
}

// 35. Count Vowels in a String in JavaScript?

function countVowels(str) {
  let count = 0;
  for (let i = 0; i < str.length; i++) {
    let ch = str[i].toLowerCase();
    if (ch === 'a' || ch === 'e' || ch === 'i' || ch === 'o' || ch === 'u') {
      count++;
    }
  }
  return count;
}
```

```
// 36. Get Unique Characters from a String in JavaScript?

function uniqueChars(str) {
  let result = "";          //declare an variable with string
  for (let i = 0; i < str.length; i++) {  // run the loop-1

    let found = false;                     // declare and intialise the
variable with false
    for (let j = 0; j < result.length; j++) {  //run the loop-2

      if (str[i] === result[j]) {             // compare loop 1 and loop 2
        found = true;                          // condition true - return true
        break;
      }
      }

// 1st for loop closed
    if (!found)
    {               //
      result += str[i];
    }
  }
  return result;
}

// 37. Find largest number and second largest number from an array?

function findTwoLargest(arr) {

  let max1 = -Infinity;
  let max2 = -Infinity;

  for (let i = 0; i < arr.length; i++) {

    if (arr[i] > max1) {   // 1st largest ->
      max2 = max1;

      max1 = arr[i];

    } else if (arr[i] > max2 && arr[i] !== max1) { //second largest
      max2 = arr[i];
    }
  }
  return [max1, max2];
}
```

```
// 38. Flatten an array without using in build flat method?

// flattern an array
// check each index , whether is number or index
//if array call that function

let arr=[5,1,50,[2,5,6],[[5]]]   //declare array
let result=[]                    //declare one variable to stroe output
function flatten(arr)            // create function
{
    for(let i in arr)            // run the for loop
    {

        if(Array.isArray(arr[i])) // check the condition whether the index is
array
        {
            flatten(arr[i])       //call the function and pass the index as
parameter- if condition is true
        }
        else
        {
            result[result.length]=arr[i]   //store the number value
        }
    }
}


flatten(arr)
console.log(result)
//print the prime numbers
let n=15

for (let j=0; j<n ; j++)
{
  if(prime(j)) // passin the function
  {
    console.log(j)
   }
}
function prime(n)
{
if(n<=1) return false
for(let i=2; i<n; i++)
{
    if(n%i==0)
    {
      return false}}
return true;
}
```

```
    </script>
</body>

</html>
```